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Option D – **Energy and the Environment**

9. *(a)* (i) The Sun has a surface temperature of 5800 K and a radius of 7.0×10^{8} m. Stating the name given to the law you use, show that the power radiated by the Sun (Solar Luminosity) is approximately 4×10^{26} W. [3]

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(ii) The main energy production mechanism in the sun is the proton-proton cycle. This consists of several fusion reactions, the net effect of which is to combine a number of protons to form one helium nucleus as shown:

$$\ldots\ldots\ldots\ {}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + 2\nu_{e} + 2\,{}^{0}_{+1}\mathrm{e}$$

I. **Complete** the equation. [1]

II. Name the particle which has the symbol ${}^{0}_{+1}\mathrm{e}$. [1]

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(iii) The energy released in the reaction is 26.7 MeV. Use this information and the answer to *(a)*(i) to determine the mean rate of production of helium nuclei in the Sun. [2]

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(b) Due to absorption in the atmosphere, the maximum intensity of the Sun's radiation received at the Earth's surface in the UK is about 750 W m^{-2}. Show that this corresponds to approximately 50 % of the solar intensity reaching the Earth's atmosphere. [Sun-Earth distance = 1.50×10^{11} m]. [2]

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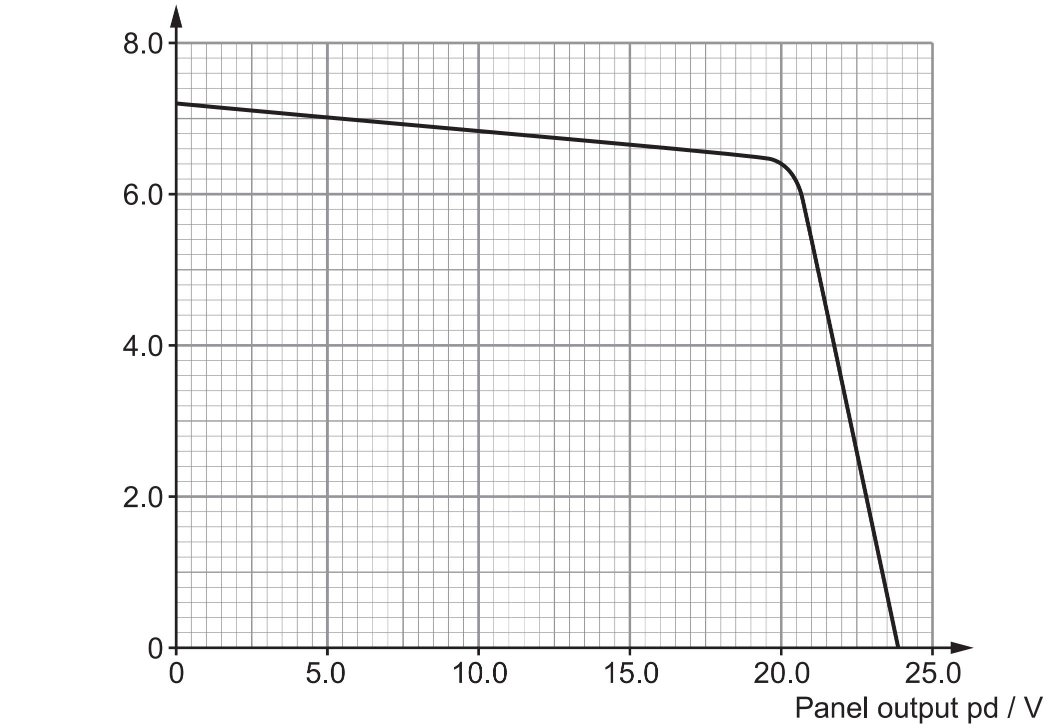
(c) Solar (PV) panels are used to produce electricity from the solar radiation incident upon them. The output power of PV panels depends on the load resistance and the intensity of the radiation. The graph shows the output characteristics for a solar panel of area 1 m^2 for varying values of load resistance for a constant **light power of 750 W**.



(i) Engineers designing this panel require that it produces at least 15 % of the maximum input power. Determine whether or not the panel meets this requirement when operating at maximum output power. [3]

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(ii) Determine the number of panels of this type needed to power a 1 kW electric kettle, and explain why, **in reality**, the actual number of panels needed will be greater. [3]

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(d) Scientists attempting to generate electricity by nuclear fusion on Earth must overcome a number of difficulties. One condition which needs to be satisfied is to ensure a high enough temperature.

(i) Explain in terms of energy and the interaction of particles why a high temperature is necessary. [3]

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(ii) For a particular nuclear fusion reaction to be successful the value of its *triple product* must be $\geq 2.6 \times 10^{28}\,\mathrm{s\,K\,m^{-3}}$. Plasma of volume $75\,\mathrm{m^3}$ contains 2.2×10^{22} reacting particles at a temperature of $120 \times 10^{6}\,\mathrm{K}$. If a confinement time of 0.8 seconds is achieved, determine whether or not fusion is possible under these conditions. [2]

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